

HOSTEL MANAGEMENT SYSTEM

Parallel and Distributed Computing Project Report

University of Central Punjab (UCP)
Faculty of Information Technology
Bachelor of Science in Artificial Intelligence (BSAI)

Course: Parallel and Distributed Computing (PDC)

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1. Abstract

This report presents the design, implementation, and performance evaluation of a Hostel Management System developed as a Parallel and Distributed Computing (PDC) lab project. The system is implemented in the C programming language using parallel arrays for data storage and provides role-based access for Admin/Warden and Student users. Core hostel operations include room management, student enrollment, fee processing, mess menu display, and complaint handling.

The primary focus of this project is the PDC Performance Test module, which benchmarks three parallel programming paradigms: Sequential execution, OpenMP multi-threading, and POSIX Threads (Pthreads). The benchmark executes a compute-intensive fee calculation workload combined with parallel student keyword search. Results recorded on an Intel Core i7-7820HQ system demonstrate speedups of up to 2.36× using OpenMP with 4 threads, validating the effectiveness of parallel computing for CPU-bound tasks.

2. Introduction

Hostel management involves coordinating rooms, student records, fee collection, mess services, and maintenance requests. Manual record-keeping is error-prone and inefficient. This project addresses these challenges by building a console-based management system that also serves as a practical platform for studying parallel programming techniques.

Parallel and Distributed Computing enables multiple processing units to work simultaneously on independent tasks, reducing total execution time. Modern processors with multiple cores can leverage OpenMP directives and POSIX thread APIs to achieve significant performance gains on suitable workloads. This project applies these concepts to a real-world hostel management scenario.

3. Problem Statement & Objectives

3.1 Problem Statement

Educational institutions require an efficient system to manage hostel operations including room allocation, fee tracking, and student services. Additionally, PDC students need hands-on experience implementing and comparing sequential versus parallel execution models on practical applications.

3.2 Objectives

- Develop a functional Hostel Management System with Admin and Student portals.
- Implement data storage using parallel arrays (no struct), as per lab requirements.
- Integrate OpenMP and POSIX Threads for parallel fee calculation and student search.
- Measure and compare execution time and speedup across 1, 2, and 4 thread configurations.
- Document system specifications and benchmark results for reproducibility.

4. System Overview

The Hostel Management System is a menu-driven console application written in C. It uses parallel arrays defined in `hostel_data.h` and seeded via `hostel_mock_data.c`. The main program (`hostel_pdc.c`) handles authentication, all business logic, and the PDC performance benchmark module.

4.1 Project Structure

File	Description
<code>hostel_pdc.c</code>	Main program — menus, logic, PDC benchmarks
<code>hostel_mock_data.c</code>	Seed/mock data initialization
<code>hostel_data.h</code>	Shared declarations and parallel array definitions
<code>search_bench.c</code>	Standalone C benchmark for web frontend

4.2 Compilation

Windows (MSYS2 / MinGW):

```
gcc -O2 -fopenmp hostel_pdc.c hostel_mock_data.c -o hostel_pdc.exe -lpthread -lm
```

Linux / Ubuntu:

```
gcc -O2 -fopenmp hostel_pdc.c hostel_mock_data.c -o hostel_pdc -lpthread -lm
```

5. Features & Functional Modules

5.1 Admin / Warden Portal

- Room Management — view, add, and delete rooms with duplicate checking.
- Student Management — view roster, search by name/roll, delete records.
- Fee Management — generate challans, record payments, view defaulters.
- Mess Menu — display weekly breakfast, lunch, and dinner schedule.
- Complaint Management — view and update complaint status.
- PDC Performance Test — run parallel benchmarks and view speedup results.

5.2 Student Portal

- View personal profile and assigned room details.
- Check fee status (rent, mess, utility dues).
- File maintenance complaints.
- View weekly mess menu.
- Access PDC Performance Test module.

5.3 Demo Credentials

Role	Username / Roll	Password
Admin / Warden	admin	admin123
Student	UCP-BSAI-1001	student123

6. Parallel Computing Implementation

The PDC module executes two combined workloads for each benchmark run: (1) Fee Risk Calculation — a compute-intensive loop using `sin()` and `cos()` over 80,000 iterations per student; and (2) Student Search — keyword matching against student names and roll numbers using `strstr()`.

6.1 Sequential Execution

The sequential baseline runs both workloads in a single thread using standard for-loops. Timing is captured via `clock()` from `time.h`, and results are stored as the $1.00\times$ speedup baseline.

6.2 OpenMP Implementation

OpenMP parallelization uses `#pragma omp parallel for` to distribute the fee calculation and student search loops across n threads (2 and 4). The `reduction(+:found)` clause is used for thread-safe accumulation of search match counts. Thread count is set via `omp_set_num_threads(n)`.

6.3 POSIX Threads (Pthreads) Implementation

The Pthread implementation manually partitions the student index range into equal chunks and spawns worker threads using `pthread_create()`. Each worker processes its assigned index range for fee calculation or search. Results are merged via `pthread_join()` and local counter arrays.

6.4 Parallel Methods Comparison

Method	Mechanism	Thread Control
Sequential	Single-threaded for-loop	1 thread (baseline)
OpenMP	<code>#pragma omp parallel for</code>	2 or 4 threads via <code>omp_set_num_threads</code>
POSIX Threads	<code>pthread_create</code> / <code>pthread_join</code>	4 explicit worker threads

7. Test Environment & System Specifications

All PDC performance benchmark results documented in this report were obtained on the following hardware and software configuration:

Specification	Value
Device Name	DESKTOP-LL6V00R
Processor	Intel(R) Core(TM) i7-7820HQ CPU @ 2.90 GHz
Installed RAM	16.0 GB (15.8 GB usable)
Graphics Card	NVIDIA GeForce 940MX (2 GB), Intel(R) HD Graphics 630 (128 MB)
Storage	384 GB of 477 GB used
System Type	64-bit operating system, x64-based processor
Base Speed	2.90 GHz
Sockets	1
Cores	4
Logical Processors	8
Virtualization	Enabled
L1 Cache	256 KB
L2 Cache	1.0 MB
L3 Cache	8.0 MB

Pen and Touch	Touch support with 10 touch points
---------------	------------------------------------

8. PDC Performance Test & Results

8.1 Test Parameters

Parameter	Value
Active Students	10
Search Keyword	Ali
Workload Iterations (WORK)	80,000 per student
CLOCKS_PER_SEC	1000
Compiler Flags	-O2 -fopenmp -lpthread -lm
Timing Function	clock() from time.h

8.2 Execution Log

- Running Sequential...
- Running OpenMP 2 threads...
- Running OpenMP 4 threads...
- Running Pthread 4 threads...

8.3 Results Table

Table 2 presents the measured execution time, CPU clock ticks, and speedup relative to the sequential baseline for each parallel method.

Method	Threads	Time (sec)	Ticks	Speedup
Sequential	1	0.0260	26	1.00x
OpenMP	2 threads	0.0120	12	2.17x
OpenMP	4 threads	0.0110	11	2.36x
Pthread (POSIX Threads)	4 threads	0.0130	13	2.00x

9. Result Analysis

The sequential execution completed in 0.0260 seconds (26 clock ticks), establishing the baseline performance. OpenMP with 2 threads achieved a speedup of 2.17 $\times$, reducing execution time to 0.0120 seconds. With 4 OpenMP threads, the best result of 2.36 $\times$ speedup was observed (0.0110 seconds), demonstrating near-linear scaling for this CPU-bound workload on a 4-core / 8-logical-processor system.

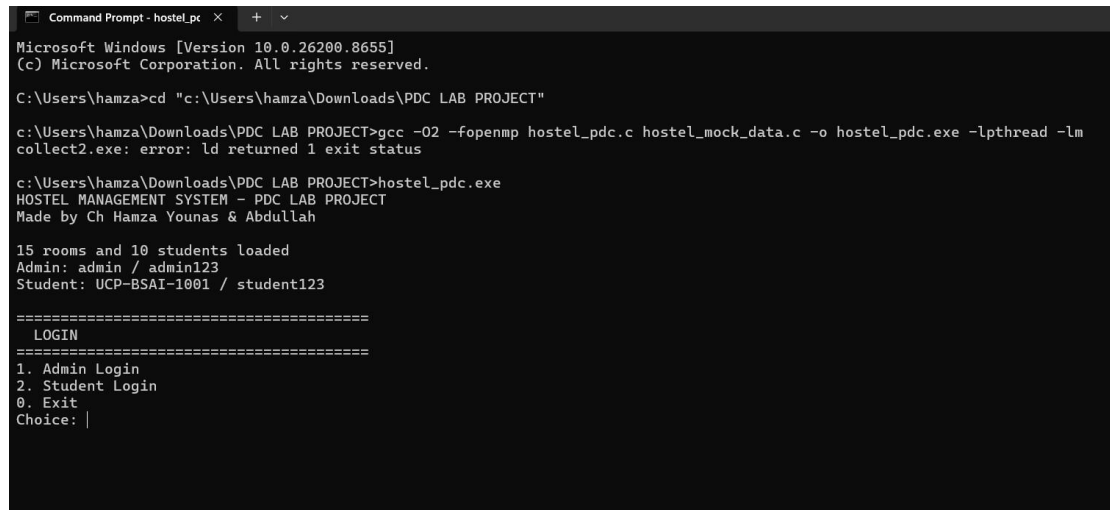
POSIX Threads with 4 workers achieved a 2.00 $\times$ speedup (0.0130 seconds). While slightly slower than OpenMP at 4 threads, Pthreads provide explicit control over thread creation, work partitioning, and synchronization — making them suitable for custom parallel designs.

OpenMP generally outperformed Pthreads in this benchmark due to lower scheduling overhead and optimized parallel loop implementation. The diminishing returns beyond 2 threads suggest

that memory bandwidth and synchronization costs begin to limit further scaling with only 10 students in the dataset.

10. Screenshots

10.1 Admin User Interface



```
Command Prompt - hostel_pc X + v
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hamza>cd "c:\Users\hamza\Downloads\PDC LAB PROJECT"

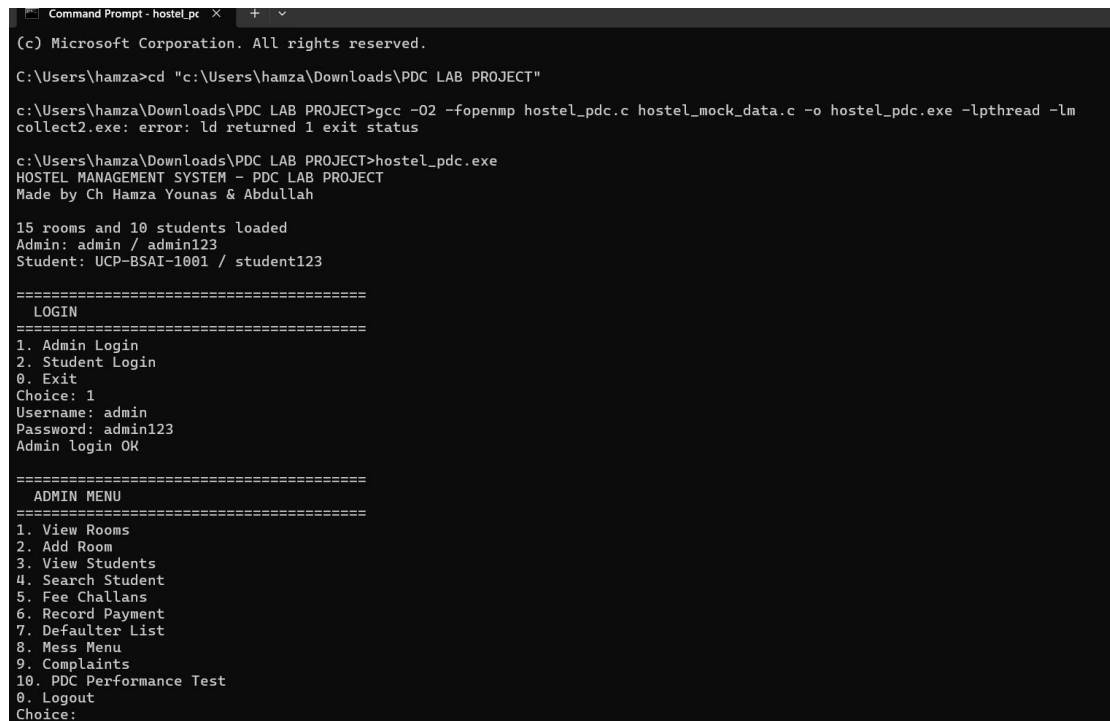
c:\Users\hamza\Downloads\PDC LAB PROJECT>gcc -O2 -fopenmp hostel_pdc.c hostel_mock_data.c -o hostel_pdc.exe -lpthread -lm
collect2.exe: error: ld returned 1 exit status

c:\Users\hamza\Downloads\PDC LAB PROJECT>hostel_pdc.exe
HOSTEL MANAGEMENT SYSTEM - PDC LAB PROJECT
Made by Ch Hamza Younas & Abdullah

15 rooms and 10 students loaded
Admin: admin / admin123
Student: UCP-BSAI-1001 / student123

=====
LOGIN
=====
1. Admin Login
2. Student Login
0. Exit
Choice: |
```

Figure 1: Admin Login Portal



```
Command Prompt - hostel_pc X + v
(c) Microsoft Corporation. All rights reserved.

C:\Users\hamza>cd "c:\Users\hamza\Downloads\PDC LAB PROJECT"

c:\Users\hamza\Downloads\PDC LAB PROJECT>gcc -O2 -fopenmp hostel_pdc.c hostel_mock_data.c -o hostel_pdc.exe -lpthread -lm
collect2.exe: error: ld returned 1 exit status

c:\Users\hamza\Downloads\PDC LAB PROJECT>hostel_pdc.exe
HOSTEL MANAGEMENT SYSTEM - PDC LAB PROJECT
Made by Ch Hamza Younas & Abdullah

15 rooms and 10 students loaded
Admin: admin / admin123
Student: UCP-BSAI-1001 / student123

=====
LOGIN
=====
1. Admin Login
2. Student Login
0. Exit
Choice: 1
Username: admin
Password: admin123
Admin login OK

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice:
```

Figure 2: Admin Main Menu

```
Command Prompt - hostel_pc  X  +  v

Username: admin
Password: admin123
Admin login OK

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 1

=====
ROOM LIST
=====
ID  Room  Floor  Type  AC  Occ/Cap  Rent
1   R-101  1      2-seater  Y  1/2      18000
2   R-102  1      4-seater  N  1/4      12000
3   R-103  1      2-seater  N  1/2      12000
4   R-104  1      4-seater  Y  1/4      18000
5   R-105  1      2-seater  N  1/2      12000
6   R-106  2      4-seater  N  1/4      12000
7   R-107  2      2-seater  Y  1/2      18000
8   R-108  2      4-seater  N  1/4      12000
9   R-109  2      2-seater  N  1/2      12000
10  R-110  2      4-seater  Y  1/4      18000
11  R-111  3      2-seater  N  0/2      12000
12  R-112  3      4-seater  N  0/4      12000
13  R-113  3      2-seater  Y  0/2      18000
14  R-114  3      4-seater  N  0/4      12000
15  R-115  3      2-seater  N  0/2      12000

Press Enter...|
```

Figure 3: Room List View

```
Command Prompt - hostel_pc  X  +  v

14 R-114 3 4-seater N 0/4 12000
15 R-115 3 2-seater N 0/2 12000
16 R-103 1 2-seater Y 0/1 5000

Press Enter...

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 2

=====
ADD ROOM
=====
Room No (R-101): R-116
Floor: 2
Type (2-seater/4-seater): 2-seater
AC (1=Yes 0=No): 1
Capacity: 3
Rent: 1000
Room added

Press Enter...
```

Figure 4: Adding a New Room

```
Command Prompt - hostel_pc  X + v
Press Enter...

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 1

=====
ROOM LIST
=====
ID Room Floor Type AC Occ/Cap Rent
1 R-101 1 2-seater Y 1/2 18000
2 R-102 1 4-seater N 1/4 12000
3 R-103 1 2-seater N 1/2 12000
4 R-104 1 4-seater Y 1/4 18000
5 R-105 1 2-seater N 1/2 12000
6 R-106 2 4-seater N 1/4 12000
7 R-107 2 2-seater Y 1/2 18000
8 R-108 2 4-seater N 1/4 12000
9 R-109 2 2-seater N 1/2 12000
10 R-110 2 4-seater Y 1/4 18000
11 R-111 3 2-seater N 0/2 12000
12 R-112 3 4-seater N 0/4 12000
13 R-113 3 2-seater Y 0/2 18000
14 R-114 3 4-seater N 0/4 12000
15 R-115 3 2-seater N 0/2 12000
16 R-103 1 2-seater Y 0/1 5000
17 R-116 2 2-seater Y 0/3 1000

Press Enter...|
```

Figure 5: Room Added Successfully

```
Command Prompt - hostel_pc  X  +  v

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 3

=====
STUDENT LIST
=====
ID  Name                Roll                Room  Rent  Mess
1   Ali Hassan           UCP-BSAI-1001      R-101  Due   Due
2   Fatima Malik         UCP-BSAI-1002      R-102  Paid  Paid
3   Usman Ahmed          UCP-BSAI-1003      R-103  Paid  Paid
4   Ayesha Khan          UCP-BSAI-1004      R-104  Due   Paid
5   Bilal Raza           UCP-BSAI-1005      R-105  Paid  Due
6   Hamza Younas         UCP-BSAI-1006      R-106  Paid  Paid
7   Abdullah Khan        UCP-BSAI-1007      R-107  Due   Paid
8   Sana Iqbal           UCP-BSAI-1008      R-108  Paid  Paid
9   Zain Butt            UCP-BSAI-1009      R-109  Paid  Due
10  Hira Nawaz           UCP-BSAI-1010      R-110  Due   Paid

Press Enter...
```

Figure 6: Student List

```
Command Prompt - hostel_pc  X  +  v

Press Enter...

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice:
4

=====
SEARCH STUDENT
=====
Enter roll or name: Hamza Younas
Hamza Younas | UCP-BSAI-1006 | Room 6

Press Enter...
```

Figure 7: Search Student by Keyword

```
Command Prompt - hostel_pc X + v

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 5

=====
FEE CHALLANS
=====
Ali Hassan | Rent:18000 Mess:8500 Utility:1500 Total:28000
Fatima Malik | Rent:12000 Mess:8500 Utility:1500 Total:22000
Usman Ahmed | Rent:12000 Mess:8500 Utility:1500 Total:22000
Ayesha Khan | Rent:18000 Mess:8500 Utility:1500 Total:28000
Bilal Raza | Rent:12000 Mess:8500 Utility:1500 Total:22000
Hamza Younas | Rent:12000 Mess:8500 Utility:1500 Total:22000
Abdullah Khan | Rent:18000 Mess:8500 Utility:1500 Total:28000
Sana Iqbal | Rent:12000 Mess:8500 Utility:1500 Total:22000
Zain Butt | Rent:12000 Mess:8500 Utility:1500 Total:22000
Hira Nawaz | Rent:18000 Mess:8500 Utility:1500 Total:28000

Press Enter...
```

Figure 8: Fee Challans

```
Command Prompt - hostel_pc X + v

ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice:
6

=====
RECORD PAYMENT
=====
Roll No: UCP-BSAI-1004
1.Rent 2.Mess 3.Both
Choice: 3
Payment saved for Ayesha Khan

Press Enter...
```

Figure 9: Record Payment

```
Command Prompt - hostel_pc  X  +  v

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 7

=====
DEFAULTER LIST
=====
Ali Hassan | Rent:Due | Mess:Due
Bilal Raza | Rent:Paid | Mess:Due
Abdullah Khan | Rent:Due | Mess:Paid
Zain Butt | Rent:Paid | Mess:Due
Hira Nawaz | Rent:Due | Mess:Paid
Total defaulters: 5

Press Enter...
```

Figure 10: Defaulter List

```
Command Prompt - hostel_pc  X  +  v

=====
MESS MENU
=====
Day 1
Breakfast: Day 1 Breakfast: Paratha + Tea
Lunch:      Day 1 Lunch: Biryani + Raita
Dinner:     Day 1 Dinner: Karahi + Naan
Day 2
Breakfast: Day 2 Breakfast: Paratha + Tea
Lunch:      Day 2 Lunch: Biryani + Raita
Dinner:     Day 2 Dinner: Karahi + Naan
Day 3
Breakfast: Day 3 Breakfast: Paratha + Tea
Lunch:      Day 3 Lunch: Biryani + Raita
Dinner:     Day 3 Dinner: Karahi + Naan
Day 4
Breakfast: Day 4 Breakfast: Paratha + Tea
Lunch:      Day 4 Lunch: Biryani + Raita
Dinner:     Day 4 Dinner: Karahi + Naan
Day 5
Breakfast: Day 5 Breakfast: Paratha + Tea
Lunch:      Day 5 Lunch: Biryani + Raita
Dinner:     Day 5 Dinner: Karahi + Naan
Day 6
Breakfast: Day 6 Breakfast: Paratha + Tea
Lunch:      Day 6 Lunch: Biryani + Raita
Dinner:     Day 6 Dinner: Karahi + Naan
Day 7
Breakfast: Day 7 Breakfast: Paratha + Tea
Lunch:      Day 7 Lunch: Biryani + Raita
Dinner:     Day 7 Dinner: Karahi + Naan

Press Enter...
```

Figure 11: Mess Menu (Admin View)

```
Command Prompt - hostel_pc X + v
Press Enter...

=====
ADMIN MENU
=====
1. View Rooms
2. Add Room
3. View Students
4. Search Student
5. Fee Challans
6. Record Payment
7. Defaulter List
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 9

=====
COMPLAINTS
=====
1 Wi-Fi not working - Pending
2 Fan not working - In Progress

Press Enter...
```

Figure 12: Complaint Management

```
Command Prompt - hostel_pc X + v
8. Mess Menu
9. Complaints
10. PDC Performance Test
0. Logout
Choice: 10

=====
PDC PERFORMANCE TEST
=====
Students: 10
CLOCKS_PER_SEC = 1000

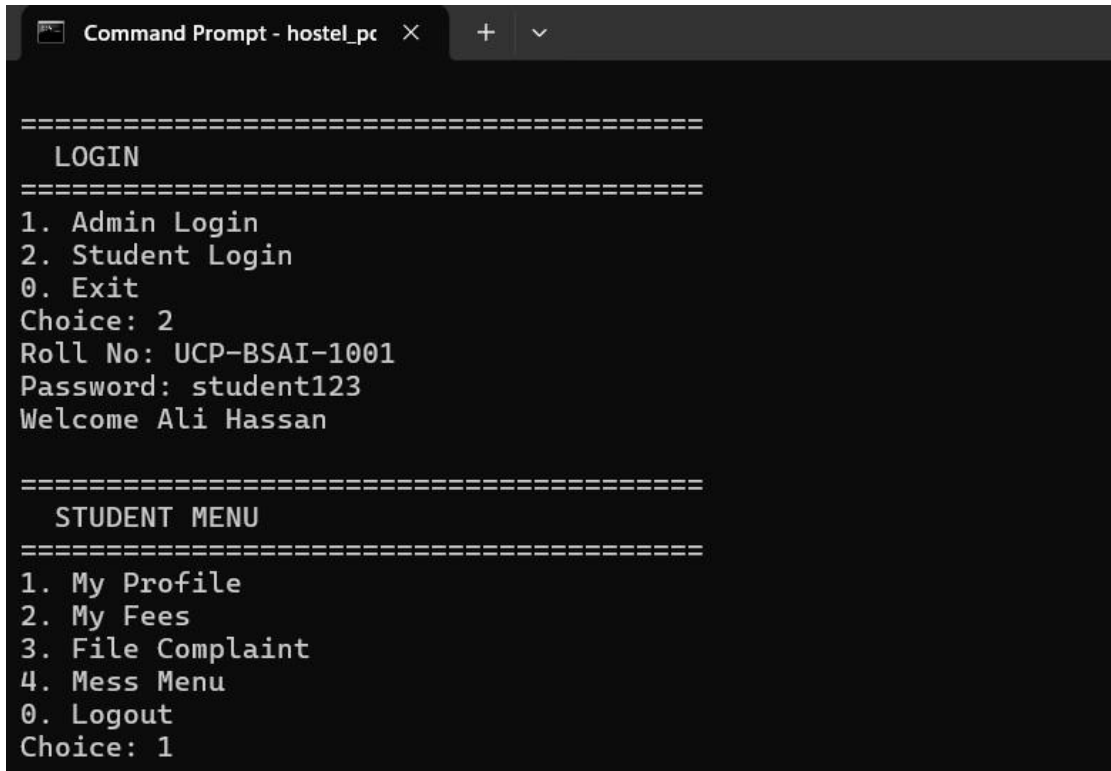
Running Sequential...
Running OpenMP 2 threads...
Running OpenMP 4 threads...
Running Pthread 4 threads...

--- RESULTS ---
Method          Time(sec)  Ticks    Speedup
Sequential      0.0260    26       1.00x
OpenMP 2 threads 0.0120    12       2.17x
OpenMP 4 threads 0.0110    11       2.36x
Pthread 4 threads 0.0130    13       2.00x

Press Enter...
```

Figure 13: PDC Performance Test (Admin)

10.2 Student User Interface



```
Command Prompt - hostel_pc  X  +  v

=====
LOGIN
=====
1. Admin Login
2. Student Login
0. Exit
Choice: 2
Roll No: UCP-BSAI-1001
Password: student123
Welcome Ali Hassan

=====
STUDENT MENU
=====
1. My Profile
2. My Fees
3. File Complaint
4. Mess Menu
0. Logout
Choice: 1
```

Figure 14: Student Main Menu

```
Command Prompt - hostel_pc  X  +  v

Welcome Ali Hassan

=====
STUDENT MENU
=====
1. My Profile
2. My Fees
3. File Complaint
4. Mess Menu
0. Logout
Choice: 1

=====
MY PROFILE
=====
Name: Ali Hassan
Roll: UCP-BSAI-1001
Phone: 0300-1000000
Room: R-101 | Floor: 1 | Type: 2-seater

Press Enter...
```

Figure 15: Student Profile

```
Command Prompt - hostel_pc X + v
Room: R-101 | Floor: 1 | Type: 2-seater
Press Enter...

=====
      STUDENT MENU
=====
1. My Profile
2. My Fees
3. File Complaint
4. Mess Menu
0. Logout
Choice: 2

=====
      MY FEES
=====
Rent:    18000 [Due]
Mess:    8500 [Due]
Utility: 1500
Total:   28000

Press Enter...
```

Figure 16: Student Fee Dashboard

```
Command Prompt - hostel_pc X + v

=====
      STUDENT MENU
=====
1. My Profile
2. My Fees
3. File Complaint
4. Mess Menu
0. Logout
Choice: 3

=====
      FILE COMPLAINT
=====
Write complaint: Garmi bht hogyi hai
Complaint submitted

Press Enter...
```

Figure 17: File a Complaint

```
Command Prompt - hostel_pc  X  +  v

=====
1. My Profile
2. My Fees
3. File Complaint
4. Mess Menu
0. Logout
Choice: 4

=====
      MESS MENU
=====
Breakfast: Day 1 Breakfast: Paratha + Tea
Lunch:      Day 1 Lunch: Biryani + Raita
Dinner:     Day 1 Dinner: Karahi + Naan
-----
Breakfast: Day 2 Breakfast: Paratha + Tea
Lunch:      Day 2 Lunch: Biryani + Raita
Dinner:     Day 2 Dinner: Karahi + Naan
-----
Breakfast: Day 3 Breakfast: Paratha + Tea
Lunch:      Day 3 Lunch: Biryani + Raita
Dinner:     Day 3 Dinner: Karahi + Naan
-----
Breakfast: Day 4 Breakfast: Paratha + Tea
Lunch:      Day 4 Lunch: Biryani + Raita
Dinner:     Day 4 Dinner: Karahi + Naan
-----
Breakfast: Day 5 Breakfast: Paratha + Tea
Lunch:      Day 5 Lunch: Biryani + Raita
Dinner:     Day 5 Dinner: Karahi + Naan
-----
Breakfast: Day 6 Breakfast: Paratha + Tea
Lunch:      Day 6 Lunch: Biryani + Raita
Dinner:     Day 6 Dinner: Karahi + Naan
-----
Breakfast: Day 7 Breakfast: Paratha + Tea
Lunch:      Day 7 Lunch: Biryani + Raita
Dinner:     Day 7 Dinner: Karahi + Naan
-----
```

Figure 18: Mess Menu (Student View)

```

Command Prompt - hostel_pc  X + v
+-----+
| PDC PERFORMANCE TEST |
+-----+
+-----+
| Students in system : 10 |
| Active students   : 10 |
| Search keyword    : "Ali" |
| CLOCKS_PER_SEC    : 1000 |
+-----+

Running parallel benchmarks...

[1/4] Sequential (1 thread)...
[2/4] OpenMP (2 threads)...
[3/4] OpenMP (4 threads)...
[4/4] Pthread (4 threads)...

+-----+
| PARALLEL BENCHMARK RESULTS |
+-----+
+-----+
| Method | Thr | Fee(s) | Srch(s) | Total(s) | Found | Speedup |
+-----+
| Sequential | 1 | 0.0240 | 0.0000 | 0.0240 | 1 | 1.00x |
| OpenMP | 2 | 0.0120 | 0.0000 | 0.0120 | 1 | 2.00x |
| OpenMP | 4 | 0.0100 | 0.0000 | 0.0100 | 1 | 2.40x |
| Pthread | 4 | 0.0130 | 0.0000 | 0.0130 | 1 | 1.85x |
+-----+

+-----+
| Method | Fee Ticks | Search Ticks | Total Ticks | Speedup |
+-----+
| Sequential | 24 | 0 | 24 | 1.00x |
| OpenMP x2 | 12 | 0 | 12 | 2.00x |
| OpenMP x4 | 10 | 0 | 10 | 2.40x |
| Pthread x4 | 13 | 0 | 13 | 1.85x |
+-----+

+-----+
| [ i ] Each run = Fee calculation + Student search |
+-----+
Keyword "Ali" matched 1 student(s) in all runs

```

Figure 19: PDC Performance Test (Student)

11. Conclusion

The Hostel Management System successfully integrates practical hostel administration features with a comprehensive parallel computing benchmark module. The project demonstrates the application of Sequential, OpenMP, and POSIX Threads programming models to a real-world compute-intensive scenario.

Benchmark results on the Intel Core i7-7820HQ platform confirm that parallel execution significantly reduces processing time, with OpenMP achieving up to 2.36× speedup. This project fulfills the PDC lab objectives of implementing, measuring, and analyzing parallel algorithms while delivering a functional management system for educational use.

We hereby declare that this project report is our own work and has been submitted for the Parallel and Distributed Computing course at UCP.